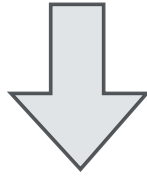
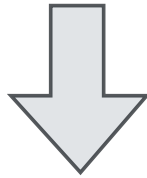


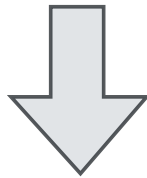
Data science Univariate



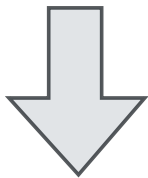
Concepts under Univariate Analysis



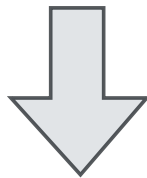
Continuous



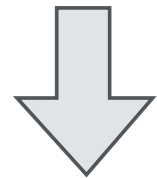
Measure of central tendency



Mean



Median



Mode

Mean median mode table for quan columns

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
Mean	108	67.3034	66.3332	66.3702	72.1006	62.2782	288655
Median	108	67	65	66	71	62	265000
Mode	1	62	63	65	60	56.7	300000

Summary of this table

Mean

- In this table can be able to measure mean values for all columns so, we got mean value in column name “ssc_p” This is average of that column, so that the students got average marks, that means very few students gets very low and high marks in ssc_p.

Median

- To find median we have to sort all values.
- We find median to know center student’s marks in every column.
- so, median 67 in “ssc_p” the middle student’s mark is 67 after sorted.
- NOTE: median omits outliers.

Mode

- We usually use mode to find repeated values in given dataset.
- “ssc_p” mode value is 62 this means that, the mark 62 is taken by most of the students.
- Note: We take mode to find high repeated values in dataset.

Measure of location of data



Percentile

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
Mean	108	67.3034	66.3332	66.3702	72.1006	62.2782	288655
Median	108	67	65	66	71	62	265000
Mode	1	62	63	65	60	56.7	300000
Q1:25%	54.5	60.6	60.9	61	60	57.945	240000
Q2:50%	108	67	65	66	71	62	265000
Q3:75%	161.5	75.7	73	72	83.5	66.255	300000
99%	212.86	87	91.86	83.86	97	76.1142	NaN
Q4:100%	215	89.4	97.7	91	98	77.89	940000

Percentile

- After sorted data we have to split as 4 quartiles.
- percentile shows range between quartiles.
- percentile of ssc_p is 60.6 this means that 60.6 is the boundary of 25% of data.
- In other words, 25% of data's end point is 60.6 (25th percentile).
- In this order we have to find rest of 3 quartiles.

Summary of this table

- I created a table with Q1,Q2,Q3 and Q4 to find percentile value for quartile 1 to 4
- Q1 values denoted 25% of data exist the range between initial to Q1 th value.
- Same we can find any percentile value.

Inter quartile range (IQR)



	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
Mean	108	67.3034	66.3332	66.3702	72.1006	62.2782	288655
Median	108	67	65	66	71	62	265000
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Q3:75%	161.5	75.7	73	72	83.5	66.255	300000
99%	212.86	87	91.86	83.86	97	76.1142	NaN
Q4:100%	215	89.4	97.7	91	98	77.89	940000
IQR	107	15.1	12.1	11	23.5	8.31	60000
1.5rule	160.5	22.65	18.15	16.5	35.25	12.465	90000
lesser_per	-106	37.95	42.75	44.5	24.75	45.48	150000
greater_per	322	98.35	91.15	88.5	118.75	78.72	390000
Mini	1	40.89	37	50	50	51.21	200000
Max	215	89.4	97.7	91	98	77.89	940000

Summary of table.

- In this table we found lesser permitted to know what least permitted value is.
- Same for greater permitted.
- Then, the row mini and max show us what is least and max value present in dataset.
- So, now we have all values next compare and replace it.

To find IQR

- $IQR = Q3 - Q1$
- $lesser_pre = Q1 - 1.5(IQR)$
- $greater_pre = Q3 + 1.5(IQR)$

How to find outliers?

How to find lesser ourlier and greater ourlier

$lesser_outlier = Mini < lesser_per$ if this condition true there is outlier

$greater_outlier = max > greater_per$ if this condition true thire is outlier.

Finally replace the outliers with permitted values.

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
Mean	108	67.3034	66.3347	66.3586	72.1006	62.2782	277649
Median	108	67	65	66	71	62	265000
Mode	1	62	63	65	60	56.7	300000
Q1:25%	54.5	60.6	60.9	61	60	57.945	240000
Q2:50%	108	67	65	66	71	62	265000
Q3:75%	161.5	75.7	73	72	83.5	66.255	300000
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lesser_per	-106	37.95	42.75	44.5	24.75	45.48	150000
greater_per	322	98.35	91.15	88.5	118.75	78.72	390000
Mini	1	40.89	42.75	50	50	51.21	200000
Max	215	89.4	91.15	88.5	98	77.89	390000

In this above table we compared and replaced outliers with permitted values.

Max of “hsc_p” 97.7 replaced with value of 91.5 because more than 91.5 is consider as a large range of outliers.

```
|: freq_Tab(dataset, "hsc_p")
```

	Unique_values	Frequency	Relative_Frequency	cumsum
0	63.00	14	0.065116	0.065116
1	62.00	12	0.055814	0.120930
2	67.00	9	0.041860	0.162791
3	60.00	9	0.041860	0.204651
4	64.00	8	0.037209	0.241860
...	...	...	...	...
86	60.80	1	0.004651	0.981395
87	89.83	1	0.004651	0.986047
88	66.60	1	0.004651	0.990698
89	87.60	1	0.004651	0.995349
90	91.00	1	0.004651	1.000000

91 rows × 4 columns

In above table we found frequency

- Frequency is repeated values of every unique values.
- Unique values starts form index “0”.
- Now we got frequency, Relative frequency and cumsum for every unique values.

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
Mean	108	67.3034	66.3347	66.3586	72.1006	62.2782	277649
Median	108	67	65	66	71	62	265000
Mode	1	62	63	65	60	56.7	300000
Q1:25%	54.5	60.6	60.9	61	60	57.945	240000
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greater_per	322	98.35	91.15	88.5	118.75	78.72	390000
Mini	1	40.89	42.75	50	50	51.21	200000
Max	215	89.4	91.15	88.5	98	77.89	390000
skew	0	-0.132649	0.162611	0.204164	0.282308	0.313576	0.8067
kurtosis	-1.2	-0.60751	0.0869008	-0.0974897	-1.08858	-0.470723	-0.239837

Summary of table

- In above table we found skewness and kurtosis.
- Skewness tells relation between mean,median and mode.
- Kurtosis tells shape and peakness.

```

#values = [value for value in range(startrange, endrange)]
#probabilities = [dist.pdf(value) for value in values]
#prob=sum(probabilities)
def tra_pdf_calculator(dataset,startrange,endrange):
    from matplotlib import pyplot
    from scipy.stats import norm
    import seaborn as sns
    ax = sns.distplot(dataset,kde=True,kde_kws={'color':'blue'},color='Green')
    #then, the following code created for plotting vertical line with red colour
    pyplot.axvline(startrange,color='Red')
    pyplot.axvline(endrange,color='Red')
    # generate a sample
    sample = dataset #(assing dataset to sample variable)
    # calculate parameters
    sample_mean =sample.mean()
    sample_std = sample.std()
    print('Mean=%.3f, Standard Deviation=%.3f' % (sample_mean, sample_std))
    # define the distribution
    dist = norm(sample_mean, sample_std)
    # sample probabilities for a range of outcomes
    values=[]
    probabilities=[]
    for value in range(startrange,endrange):
        values.append(value)
        probabilities.append(dist.pdf(value))
    #values = [value for value in range(startrange, endrange)]
    #probabilities = [dist.pdf(value) for value in values]
    prob=sum(probabilities)
    print("The area between range({},{}):{}".format(startrange,endrange,sum(probabilities)))
    return prob

```

```
tra_pdf_calculator(dataset["ssc_p"],56,77)
```

Mean=67.303, Standard Deviation=10.827

The area between range(56,77):0.664518247929439

Explanation of this code

- In above table we found probability density function.
- we have to find PDF for all columns one by one
- so we just write and convert our code into function for future use.
- values=[] (we create empty lists to store values.)
- probabilities=[]
- for value in range(startrange,endrange): #(in above function we just give input start range and end range.)
- values.append(value) #(These inputs went and back into for loop with list of values and probabilities.)
- probabilities.append(dist.pdf(value)) (In this code our list of values calculated for probability density function)


```
def std_nb_plot(dataset): # Coverted to standard Normal Distribution

    import seaborn as sns
    mean=dataset.mean()
    std=dataset.std()

    values=[i for i in dataset] #we pass dataset to convert as list so,we can take any column for output

    z_score=[((j-mean)/std) for j in values] #we know formulae z_score= (x-mean / std)

    sns.distplot(z_score,kde=True) #now plot z_score distribution

    sum(z_score)/len(z_score)
    return z_score
```

Explanation of this code

- To convert Normal distribution into standard normal distribution we have to create funtion.
- formulae = $Z_score = \frac{x - \text{mean}}{\text{standard deviation}}$.
- **values=[i for i in dataset]** (#we pass dataset to convert as list so,we can take any column for output)
- **z_score=[((j-mean)/std) for j in values]** #we know formulae $z_score = \frac{(x - \text{mean})}{\text{std}}$
-
- **sns.distplot(z_score,kde=True)** #now plot z_score distribution

Thank You

